Week 14 Assignment

When assumptions fail, and choosing a model

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 15. See the syllabus for the late policy.

```
library(tidyverse)
library(marginaleffects)
```

Part A: Ranks. Using `chickwts` (weight by feed type), restrict to `casein` and `sunflower`.

1. Run `wilcox.test()` (use `exact = FALSE`).
2. Run the same comparison with `lm(rank(weight) ~ feed)`. Compare conclusions.

Part B: Robust inference. Using all of `chickwts`:

3. Fit `lm(weight ~ feed)`. Use `avg_comparisons()` for comparisons to the reference feed, with the default and with `vcov = "HC3"`. Which comparisons change the most?

Part C: Choosing a model.

4. For each scenario give the response type, the predictors, the independence structure, and the `R` call you'd use (a formula is enough):
   a. Counts of aphids on 30 plants, half sprayed, half not.
   b. Heart rate before and after caffeine in 15 students.
   c. Presence/absence of a species in 50 ponds across 5 lakes, as a function of pond area.
5. In 3–4 sentences: how has "everything is a linear model" changed how you think about picking a statistical test?